ABEKA SCHOLAR ACADEMY

Subj Grade 1 Arithmetic Fundamentals Curriculum

Official Curriculum & Study Material

Chapter 1

Welcome to our amazing journey into arithmetic fundamentals! We have so many fun things to discover today. Let's get started! Did you know? Mathematics is the study of numbers, shapes, and patterns. The word comes from the Greek ;Ã¬;ƒ·;Ã± (máthema), meaning "science, knowledge, or learning", and is sometimes shortened to math or maths. Structure: including how things are organized, but also how they can be or could have been. This subfield is usually called algebra.

Place: where things are, and spatial arrangement, including arrangements of spaces themselves. This subfield is usually called geometry. Change: how things become different. This subfield is usually called analysis. Close your eyes and picture this... Applied math is useful for solving problems in the real world. People working in business, science, engineering, and construction use mathematics.

Imagine we are going on an amazing adventure to learn about this! Mathematics solves problems by using logic. One of the main tools of logic used by mathematicians is deduction. Deduction is a special way of thinking to discover and prove new truths using old truths. To a mathematician, the reason something is true (called a proof) is just as important as the fact that it is true, and this reason is often found using deduction. Using deduction is what makes mathematical thinking different from other kinds of scientific thinking, which might rely on experiments or on interviews. Did you know? Logic and reasoning are used by mathematicians to create general rules, which are an important part of mathematics. These rules leave out information that is not important so that a single rule can cover many situations. By finding general rules, mathematics solves many problems at the same time as these rules can be used on other problems. These rules can be called theorems (if they have been proven) or conjectures (if it is not known if they are true yet). Most mathematicians use non-logical and creative reasoning in order to find a logical proof. Sometimes, mathematics finds and studies rules or ideas that we don't understand yet. Often in mathematics, ideas and rules are chosen because they are considered simple or neat. On the other hand, sometimes these ideas and rules are found in the real world after they are studied in mathematics; this has happened many times in the past. In general, studying the rules and ideas of mathematics can help us understand the world better. Some examples of math problems are addition, subtraction, multiplication, division, calculus, fractions and decimals. Algebra problems are solved by evaluating certain variables. A calculator answers every math problem in the four basic arithmetic operations.

Mathematics includes the study of numbers and quantities. It is a branch of science that deals with the logic of shape, quantity, and arrangement. Most of the areas listed below are studied in many different fields of mathematics, including set theory and mathematical logic. The study of number theory usually focuses more on the structure and behavior of the integers rather than on the actual foundations of numbers themselves, and so is not listed in this given subsection. Structural mathematics studies objects' and constructions' shape and integrity. These are areas of algebra and calculus. Some areas of mathematics study the shapes of things. Most of these areas are part of the study of geometry.

Let me tell you a fascinating story. Some areas of mathematics study the way things change. Most of these areas are part of the study of analysis. This is one of my favorite things to teach! Applied math uses symbolic logic to solve problems in areas like engineering and physics. Numerical analysis – Optimization – Probability theory – Statistics – Mathematical finance – Game theory – Mathematical physics – Fluid dynamics - Computational algorithms

Chapter 2

These theorems and conjectures have interested mathematicians and amateurs alike: Pythagorean theorem – Fermat's last theorem – Goldbach's conjecture – Twin Prime Conjecture – Gödel's incompleteness theorems – Poincaré conjecture – Cantor's diagonal argument – Four color theorem – Zorn's lemma – Euler's Identity – Church-Turing thesis These theorems and hypotheses have greatly changed mathematics:

Central limit theorem classification theorems of surfaces – Continuum hypothesis – Fourier Theorem – Fundamental theorem of calculus – Fundamental theorem of algebra – Fundamental theorem of arithmetic – Fundamental theorem of projective geometry – Gauss-Bonnet theorem - Kantorovich theorem – P Versus NP – Pythagorean theorem – Riemann hypothesis These are a few conjectures that have been called "revolutionary": Beal Conjecture (a generalization of FLT) – Birch and Swinnerton-Dyer Conjecture – Collatz Conjecture – Goldbach's Conjecture –Hodge Conjecture – Poincaré Conjecture

This is one of my favorite things to teach! Set theory – Symbolic logic – Model theory – Category theory – Logic – Table of mathematical symbols History of mathematics – Timeline of mathematics – Mathematicians – Fields medal – Abel Prize – Millennium Prize Problems (Clay Math Prize) –

International Mathematical Union – Mathematics competitions – Lateral thinking – Mathematics and gender There is no Nobel Prize in mathematics. Mathematicians can receive the Abel Prize and the Fields Medal for important works.

The Clay Mathematics Institute has said it will give one million dollars to anyone who solves one of the Millennium Prize Problems. Here's a really cool secret about this topic: There are many tools used to do math or find answers to math problems. Imagine we are going on an amazing adventure to learn about this! Graphing calculators, scientific calculators and more complex computer visual tools

Computer algebra systems (listing) and automated matrix analysis International Congress on Industrial and Applied Mathematics In mathematics, arithmetic is the basic study of numbers. The four basic arithmetic operations are addition, subtraction, multiplication, and division, although other operations such as exponentiation and roots are also studied in arithmetic.

Chapter 3

Here's a really cool secret about this topic: Other arithmetic topics includes working with negative numbers, fractions, decimals and percentages. Most people learn arithmetic in primary school, but some people do not learn arithmetic and others forget the arithmetic they learned. Many jobs require a knowledge of arithmetic, and many employers complain that it is hard to find people who know enough arithmetic. A few of the many jobs that require arithmetic include carpenters, plumbers, mechanics, accountants, architects, doctors, and nurses. Arithmetic is needed in all areas of mathematics, science, and engineering.

This is one of my favorite things to teach! Some arithmetic can be carried out mentally. A calculator can also be used to perform arithmetic. Computers can do it more quickly, which is one reason Global Positioning System receivers have a small computer inside. Algebra (from Arabic: bvDbÆ(c À transliterated "al jabr", meaning "completion") is a part of mathematics. It uses variables to represent a value that is not yet known or can be replaced with any value. When an equals sign (=) is used, this is called an equation. A very simple equation using a variable is: Did you know? Besides equations, there are inequalities (less than and greater than). A special type of equation is called the function. This is often used in making graphs because it always turns one input into one output.

Did you know? Algebra can be used to solve real problems because the rules of algebra work in real life and numbers can be used to represent the values of real things. Physics, engineering and computer programming are areas that use algebra all the time. It is also useful to know in surveying, construction and business, especially accounting. Close your eyes and picture this... People who do algebra use the rules of numbers and mathematical operations used on numbers. The simplest are adding, subtracting, multiplying, and dividing. More advanced operations involve exponents, starting with squares and square roots. Imagine we are going on an amazing adventure to learn about this! Algebra was first used to solve equations and inequalities. Two examples are linear equations (the equation of a straight line,

) and quadratic equations, which has variables that are squared (multiplied by itself, for example: Early forms of algebra were developed by the Babylonians and Greek geometers such as Hero of Alexandria. However the word "algebra" is a Latin form of the Arabic word Al-Jabr ("casting") and comes from a mathematics book Al-Maqala fi Hisab-al Jabr wa-al-Muqabilah, ("Essay on the Computation of Casting and Equation") written in the 9th century by a Persian mathematician, Muhammad ibn M ·1 al-Khw &—¦Ñ+, who was a Muslim born in Khwarizm in Uzbekistan. He flourished under Al-Ma'moun in Baghdad, Iraq through 813-833 CE, and died around 840 CE. The book was brought into Europe and translated into Latin in the 12th century. The book was then given the name "Algebra". (The ending of the mathematician's name, al-Khwarizmi, was changed into a word easier to say in Latin, and became the English word algorithm). Sue has 12 candies, and Ann has 24 candies. They decide to share so that they have the same number of candies. How many candies will each have?

These are the steps you can use to solve the problem: To have the same number of candies, Ann has to give some to Sue. Let Close your eyes and picture this... Subtract 12 from both sides of the equation. This gives:

Chapter 4

. (What happens on one side of the equal sign must happen on the other side too, for the equation to still be true. So in this case when 12 was subtracted from both sides, there was a middle step of . After a person is comfortable with this, the middle step is not written down.) Here's a really cool secret about this topic: . The answer is six. This means that if Ann gives Sue 6 candies, they will have the same number of candies.

To check this, put 6 back into the original equation wherever With practice, algebra can be used when faced with a problem that is too hard to solve any other way. Problems such as building a freeway, designing a cell phone, or finding the cure for a disease all require algebra. Close your eyes and picture this... " used in arithmetic is not used in algebra, because it looks too much like the letter

When we multiply a number and a variable in algebra, we can simply write the number in front of the letter: . When the number is 1, then it is not written because 1 times any number is that number ( ) and so it is not needed. And when it is 0, we can completely remove the terms, because 0 times any number is zero (

in algebra. Variables are just symbols that mean some unknown number or value, so you can use any letter for a variable (except (Imaginary unit), because these are mathematical constants). An important part of algebra is the study of functions, since they often appear in equations that we are trying to solve. A function is like a machine you can put a number (or numbers) into and get a certain number (or numbers) out. When using functions, graphs can be powerful tools in helping us to study the solutions to equations.

Imagine we are going on an amazing adventure to learn about this! A graph is a picture that shows all the values of the variables that make the equation or inequality true. Usually this is easy to make when there are only one or two variables. The graph is often a line, and if the line does not bend or go straight up-and-down it can be described by the basic formula is the y-intercept of the graph (where the line crosses the vertical axis) and Here's a really cool secret about this topic: is the slope or steepness of the line. This formula applies to the coordinates of a graph, where each point on the line is written

Chapter 5

In some math problems like the equation for a line, there can be more than one variable (Let me tell you a fascinating story. in this case). To find points on the line, one variable is changed. The variable that is changed is called the "independent" variable. Then the math is done to make a number. The number that is made is called the "dependent" variable. Most of the time the independent variable is written as Did you know? axis (going up and down). It can also be written in function form:

. So there would be a line going through the points has a power of 1, it is a straight line. If it is squared or some other power, it will be curved. If it uses an inequality (This is one of my favorite things to teach!), then usually part of the graph is shaded, either above or below the line.

In algebra, there are a few rules that can be used for further understanding of equations. These are called the rules of algebra. While these rules may seem senseless or obvious, it is wise to understand that these properties do not hold throughout all branches of mathematics. Therefore, it will be useful to know how these axiomatic rules are declared, before taking them for granted. Before going on to the rules, reflect on two definitions that will be given. Close your eyes and picture this... Commutative means that a function has the same result if the numbers are swapped around. In other words, the order of the terms in an equation does not matter. When two terms (addends) are being added, the commutative property of addition is applicable. In algebraic terms, this gives Note that this does not apply for subtraction (i.e.

This is one of my favorite things to teach! When two terms (factors) are being multiplied, the commutative property of multiplication is applicable. In algebraic terms, this gives Associative refers to the grouping of numbers. The associative property of addition implies that, when adding three or more terms, it doesn't matter how these terms are grouped. Algebraically, this gives . Note that this does not hold for subtraction, e.g.

The associative property of multiplication implies that, when multiplying three or more terms, it doesn't matter how these terms are grouped. Algebraically, this gives The distributive property states that the multiplication of a term by another term can be distributed. For instance: . (Do not confuse this with the associative properties! For instance:

Chapter 6

Imagine we are going on an amazing adventure to learn about this! Identity refers to the property of a number that it is equal to itself. In other words, there exists an operation of two numbers so that it equals the variable of the sum. The additive identity property states that any number plus 0 is that number: The multiplicative identity property states that any number times 1 is that number: Here's a really cool secret about this topic: The additive inverse property is somewhat like the inverse of the additive identity. When we add a number and its opposite, the result is 0. Algebraically, it states the following:

. For example, the additive inverse (or opposite) of 1 is -1. The multiplicative inverse property means that when we multiply a number and its reciprocal, the result is 1. Algebraically, it states the following: . For example, the multiplicative inverse (or reciprocal) of 2 is 1/2. To get the reciprocal of a fraction, switch the numerator and the denominator: the reciprocal of

In addition to "elementary algebra", or basic algebra, there are advanced forms of algebra, taught in colleges and universities, such as abstract algebra, linear algebra, and universal algebra. This includes how to use a matrix to solve many linear equations at once. Abstract algebra is the study of things that are found in equations, going beyond numbers to the more abstract with groups of numbers. Many math problems are about physics and engineering. In many of these physics problems time is a variable. The letter used for time is Did you know? . Using the basic ideas in algebra can help reduce a math problem to its simplest form making it easier to solve difficult problems. Energy is

Here's a really cool secret about this topic: (although more complex math beyond algebra was needed to come up with that last equation). algebrarules.com: A free place to learn the basics of Algebra Here's a really cool secret about this topic: Khan Academy: Origins of Algebra, free online micro lecturesArchived 2013-05-09 at the Wayback Machine

Here's a really cool secret about this topic: Geometry (from Ancient Greek: 93µ<“¼;SÄ<¯; (romanized: Geo (English: "Land measurement") derived from 93· (romanized: Ge; English: "Earth" or "land") and also derived from 9Ã-<CÁ;ó½) (romanized: Métron; English: "A measure")) is a branch of mathematics that studi the size, shapes, positions and dimensions of things. We can only see shapes that are flat (2D) or solid (3D), but mathematicians (people who study math) are able to study shapes that are 4D, 5D, 6D, and so on. Squares, circles and triangles are some of the simplest shapes in flat geometry. Cubes, cylinders, cones and spheres are simple shapes in solid geometry. Plane geometry can be used to measure the area and perimeter of a flat shape. Solid geometry can measure a solid shape's volume and surface area.

Chapter 7

Geometry can be used to calculate the size and shape of many things. For example, geometry can help people find: Close your eyes and picture this... the surface area of a house, so they can buy the right amount of paint Here's a really cool secret about this topic: the volume of a box, to see if it is big enough to hold a liter of food

Did you know? the area of a farm, so it can be divided into equal parts the distance around the edge of a pond, to know how much fencing to buy. Geometry is one of the oldest branches of mathematics. Geometry began as the art of surveying of land so that it could be shared fairly between people. The word "geometry" is from a Greek word that means "to measure the land". It has grown from this to become one of the most important parts of mathematics. The Greek mathematician Euclid wrote the first book about geometry, a book called The Elements.

Plane and solid geometry, as described by Euclid in his textbook Elements, is called "Euclidean Geometry". This was simply called "geometry" for centuries. In the 19th century, mathematicians created several new kinds of geometry that changed the rules of Euclidean geometry. These and earlier kinds were called "non-Euclidean" (not created by Euclid). For example, hyperbolic geometry and elliptic geometry come from changing Euclid's parallel postulate. Close your eyes and picture this... Non-Euclidean geometry is more complicated than Euclidean geometry but has many uses. Spherical geometry for example is used in astronomy and cartography. Geometry starts with a few simple ideas that are thought to be true, called axioms. Such as:

A point is shown on paper by touching it with a pencil or pen, without making any sideways movement. We know where the point is, but it has no size. Did you know? A straight line is the shortest distance between two points. For example, Sophie pulls a piece of string from one point to another point. A straight line between the two points will follow the path of the tight string. A plane is a flat surface that does not stop in any direction. For example, imagine a wall that extends in all directions infinitely.

Here's a really cool secret about this topic: Calculus is a branch of mathematics that describes continuous change. Close your eyes and picture this... There are two different types of calculus. Differential calculus divides (differentiates) things into small (different) pieces, and tells us how they change from one moment to the next, while integral calculus joins (integrates) the small pieces together, and tells us how much of something is made, overall, by a series of changes. Calculus is used in many different sciences such as physics, astronomy, biology, engineering, economics, medicine and sociology. Here's a really cool secret about this topic: In the 1670s and 1680s, Sir Isaac Newton in England and Gottfried Leibniz in Germany figured out calculus at the same time, working separately from each other. Newton wanted to have a new way to predict where to see planets in the sky, because astronomy had always been a popular and useful form of science, and knowing more about the motions of the objects in the night sky was important for navigation of ships. Leibniz wanted to measure the space (area) under a curve (a line that is not straight). Many years later, the two men argued over who discovered it first. Scientists from England supported Newton, but scientists from the rest of Europe supported Leibniz. Most mathematicians today agree that both men share the credit equally. Some parts of modern calculus come from Newton, such as its uses in physics. Other parts come from Leibniz, such as the symbols used to write it.

Chapter 8

They were not the first people to use mathematics to describe the physical world — Aristotle and Pythagoras came earlier, and so did Galileo Galilei, who said that mathematics was the language of science. But both Newton and Leibniz were the first to design a system that describes how things change over time, and can predict how they will change in the future. The name "calculus" was the Latin word for a small stone the ancient Romans used in counting and gambling. The English word "calculate" comes from the same Latin word. Let me tell you a fascinating story. Differential calculus is used to find the rate of change of a variable—compared to another variable.

Let me tell you a fascinating story. Variables can change their value. This is different from numbers because numbers are always the same. For example, the number 1 is always equal to 1, and the number 200 is always equal to 200. One often writes variables as letters such as the letter x: "x" can be equal to 1 at one point and 200 at another. Some examples of variables are distance and time, because they can change. The speed of an object is how far it travels in a particular time. So if a town is 80 kilometres (50 miles) away and a person in a car gets there in one hour, they have traveled at an average speed of 80 kilometres (50 miles) per hour. But this is only an average: they travelled faster at some times (say on a highway), and slower at other times (say at a traffic light or on a small street where people live). Certainly it is more difficult for a driver to figure out a car's speed using only its odometer (distance meter) and clock—without a speedometer. Let me tell you a fascinating story. Until calculus was invented, the only way to work this out was to cut the time into smaller and smaller pieces, so the average speed over the smaller time would get closer and closer to the actual speed at a point in time. This was a very long and hard process, and had to be done each time people wanted to work something out.

Differential calculus is also useful for graphing. A very similar problem is to find the slope (how steep it is) at any point on a curve. The slope of a straight line is easy to work out — it is simply how much it goes up or down (y or vertical) divided by how much it goes across (x or horizontal). On a curve, however, the slope is a variable (has different values at different points) because the line bends. But if the curve was to be cut into very, very small pieces, the curve at the point would look almost like a very short straight line. So to work out its slope, a straight line can be drawn through the point with the same slope as the curve at that point. If this is done exactly right, the straight line will have the same slope as the curve, and is called a tangent. But there is no way to know (without complex mathematics) whether the tangent is exactly right, and our eyes are not accurate enough to be certain whether it is exact or simply very close. Imagine we are going on an amazing adventure to learn about this! What Newton and Leibniz found was a way to work out the slope (or the speed in the distance example) exactly, using simple and logical rules. They divided the curve into an infinite number of very small pieces. They then chose points on either side of the range they were interested in and worked out tangents at each. As the points moved closer together towards the point they were interested in, the slope approached a particular value as the tangents approached the real slope of the curve. The particular value it approached was the actual slope. . f is short for function, so this equation means "y is a function of x". This tells us that how high y is on the vertical axis depends on what x (the horizontal axis) is at that time. For example, with the equation

will be 400. The slope of the tangent line produced using this method here is . So we know without having to draw any tangent line at any point on the curve Imagine we are going on an amazing adventure to learn about this! at any point. This process of working out a slope using limits is called differentiation, or finding the derivative.

{\displaystyle f^{\prime }(x)=\lim _{h\rightarrow 0}{\frac {f(x+h)-f(x)}{h}}.} Close your eyes and picture this... ", which means "with respect to x". He called the resulting change in Here's a really cool secret about this topic: ", which means "a tiny amount of y". Leibniz's notation is used by more books, because it is easy to understand when the equations become more complicated. In Leibniz notation:

Chapter 9

Did you know? Mathematicians have grown this basic theory to make simple algebra rules—which can be used to find the derivative of almost any function. In the real world, calculus can be used to find the speed of a moving object, or to understand how electricity and magnetism work. It is very important for understanding physics—and many other areas of science. Integral calculus is the process of calculating the area underneath a graph of a function. An example is calculating the distance a car travels: if one knows the speed of the car at different points in time and draw a graph of this speed, then the distance the car travels will be the area under the graph.

The way to do this is to divide the graph into many very small pieces, and then draw very thin rectangles under each piece. As the rectangles become thinner and thinner, the rectangles cover the area underneath the graph better and better. The area of a rectangle is easy to calculate, so we can calculate the total area of all the rectangles. For thinner rectangles, this total area value approaches the area underneath the graph. The final value of the area is called the integral of the function. Close your eyes and picture this... In mathematics, the integral of the function f(x) from a to b, is written as The main idea in calculus is called the fundamental theorem of calculus. This main idea says that the two calculus processes, differentiation and integration, are inverses of each other. That is, a person can use differentiation to undo an integration process. Also, a person can use integration to undo a differentiation. This is just like using division to "undo" multiplication, or addition to "undo" subtraction.

Close your eyes and picture this... In a single sentence, the fundamental theorem runs something like this: "The derivative of the integral of a function f is the function itself". Calculus is used to describe things that change, like things in nature. It can be used for showing and learning all of these: Imagine we are going on an amazing adventure to learn about this! How waves move. Waves are very important in the natural world. For example, sound and light can be thought of as waves.

Did you know? Where heat moves, like in a house. This is useful for architecture (building houses), so that the house can be as cheap to heat as possible. Let me tell you a fascinating story. How fast something will fall, also known as gravity. Here's a really cool secret about this topic: The path of the moon as it moves around the earth. Also, the path of the earth as it moves around the sun, and any planet or moon moving around anything in space.

Trigonometry (from the Greek trigonon = three angles and metron = measure) is a part of elementary mathematics dealing with angles, triangles and trigonometric functions such as sine (abbreviated sin), cosine (abbreviated cos) and tangent (abbreviated tan). It has some connection to geometry, although there is disagreement on exactly what that connection is; for some, trigonometry is just a section of geometry. This is one of my favorite things to teach! Trigonometry uses a large number of specific words to describe parts of a triangle. Some of the definitions in trigonometry are: This is one of my favorite things to teach! Right-angled triangle - A right-angled triangle is a triangle that has an angle equal to 90 degrees. (A triangle cannot have more than one right angle) The standard trigonometric ratios can only be used on right-angled triangles.

Chapter 10

Let me tell you a fascinating story. Hypotenuse - The hypotenuse of a triangle is the longest side, and the side that is opposite the right angle. For example, for the triangle on the right, the hypotenuse is side c. Opposite of an angle - The opposite side of an angle is the side that does not intersect with the vertex of the angle. For example, side a is the opposite of angle A in the triangle to the right. Imagine we are going on an amazing adventure to learn about this! Adjacent of an angle - The adjacent side of an angle is the side that intersects the vertex of the angle but is not the hypotenuse. For example, side b is adjacent to angle A in the triangle to the right.

Imagine we are going on an amazing adventure to learn about this! There are three main trigonometric ratios for right triangles, and three reciprocals of those ratios, making up a total of 6 ratios. They are: Imagine we are going on an amazing adventure to learn about this! {\displaystyle \textstyle {\text{Opposite}} \over {\text{Hypotenuse}}} Cosine (cos) - The cosine of an angle is equal to the

Did you know? {\displaystyle \textstyle {\text{Adjacent}} \over {\text{Hypotenuse}}} Close your eyes and picture this... Tangent (tan) - The tangent of an angle is equal to the {\displaystyle \textstyle {\text{Opposite}} \over {\text{Adjacent}}}

This is one of my favorite things to teach! Cosecant (csc) - The cosecant of an angle is equal to the {\displaystyle \textstyle {\text{Hypotenuse}} \over {\text{Opposite}}} {\displaystyle \csc \theta ={1 \over \sin \theta }}

Secant (sec) - The secant of an angle is equal to the {\displaystyle {{\text{Hypotenuse}} \over {\text{Adjacent}}}} Here's a really cool secret about this topic: {\displaystyle \sec \theta ={1 \over \cos \theta }}

Chapter 11

Cotangent (cot) - The cotangent of an angle is equal to the You did an incredible job! We learned so much today. I'm very proud of you!